

QINGYU ZHANG (Alfred)

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Portfolio: <https://qingyuzhang.com/>

Summary

I am a PhD researcher at University College London (UCL), funded by the EPSRC and China Scholarship Council (CSC), specialising in bioelectronic and robotic systems for intelligent medical intervention. My research focuses on integrating multimodal sensing, bioelectronic interfaces, haptics, and learning-based control to enable safe and adaptive robotic interaction with complex physical environments. Before joining UCL, I graduated top of my MSc class with Distinction from the University of Bristol and worked in IoT and intelligent hardware development, co-inventing six patents.

My broader research interests include **bioelectronic system design, medical robotics, tactile and multimodal sensing, reinforcement learning for robotics, embodied intelligence, and vision-language-action models.

EDUCATION

<i>PhD in Electronic and Electrical Engineering, UCL</i>	Mar 2023 – Now
<i>MSc in Advanced Microelectronic System Engineering (AMSE), UoB</i>	Sep 2022 - Sep 2023
- Degree Class: Distinction (1st place in the class of years 22-23, overall score 83/100) - Final project mark: 88/100	
<i>BEng in Optical Electronic Information Engineering, Huazhong University of Science and Technology</i>	Sep 2016 - Jul 2020
- Overall grade: GPA 3.22/4 - Final project mark: 88/100	

WORK EXPERIENCE

<i>Senior IoT Engineer, New Oriental Education Technology Group (XDF), Beijing, China</i>	Feb 2021 - Dec 2021
- Engineered a scalable and configurable full-stack controller on the ESP32 platform for smart classroom systems. Implemented BLE, audio, IMU, and multi-button peripheral integration; contributed to manufacturing of 200+ units. - Led the development of an AI-enabled edge device for smart classroom instruction, integrating embedded sensing, on-device model inference, and real-time feedback. Supervised three IoT engineers in system design, ML workflow implementation, and modular firmware development.	
<i>IoT Engineer, Shensi Shudun Technology, Co, Ltd, Beijing, China</i>	Jul 2020 - Jan 2021
- Developed Windows-based desktop applications using PyQt6 to streamline and automate assembly-line operations. - Led the design and implementation of a smart dock for existing air purifiers on STM32 and ESP8266 platforms, featuring one-click network configuration and multi-protocol communication (Wi-Fi, UART, I²C); the system is mass-produced with 1,000+ units annually. - Optimized the hardware communication architecture on the M263KI chip by extending SPI communication from four to six serial channels, significantly improving data throughput and system efficiency.	

SKILLS

- Coding Skills:** C/C++ (8/10), Python (7/10), Matlab (4/10), Verilog (5/10), Typescript (2/10)
- Embedded Systems:** Proficient in development across Linux, embedded RTOS, and Android platforms
- 3D Design & Simulation:** Skilled in SolidWorks, Blender, and COMSOL for 3D modelling and simulation, with specialized experience in co-simulating electrical properties, particularly electric field analysis for bioimpedance
- Circuit Design:** Experienced in PCB schematic design, layout optimisation, and hardware debugging across power management systems, embedded system integration, mixed-signal (analog/digital) circuits, and application-specific integrated circuits
- Machine Learning Skills:** PyTorch (8/10), with experience in computer vision (e.g., segmentation, classification) and reinforcement learning
- Robotic Platform:** Hands-on experience with UR collaborative robots, haptic devices (Haply Inverse3), and custom-built robotic prototypes
- Research and Research Writing Skills**

PUBLICATIONS

Paper:

- **Q. Zhang**, A. Al-Hindawi, A. Demosthenous, and Y. Wu, “Bioimpedance-Informed Haptic Needle System with Human-in-the-Loop Control for Central Venous Access,” *IEEE/ASME Transactions on Mechatronics (TMECH)*, under review.
- J. Li, D. Jiang, N. Neshatvar, **Q. Zhang**, Y. Wu, and A. Demosthenous, “A Neural Stimulator ASIC With an Improved Blocking Capacitor Charge Balancing Method and Speed-Boosting Feedback,” *IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-I)*, 2026.

Conference Paper:

- J. Li, **Q. Zhang**, D. Jiang, S. Ha, A. Demosthenous, and Y. Wu, “A 0.62 μ W/sensor 82 fps Time-to-Digital Impedance Measurement IC with Unified Excitation/Readout Front-End for Large-Scale Piezo-Resistive Sensor Array,” *IEEE International Solid-State Circuits Conference (ISSCC)*, 2026.
- **Q. Zhang**, A. Al-Hindawi, A. Demosthenous, and Y. Wu, “Haptic-Enhanced Bioimpedance Needle for Precision Navigation in Central Venous Catheterisation with Millimetre Accuracy,” *IEEE International Symposium on Circuits and Systems (ISCAS)*, 2025.
- **Q. Zhang**, A. Al-Hindawi, A. Demosthenous, and Y. Wu, “Live Demonstration: Haptic-Enhanced Bioimpedance Needle for Assisting Central Venous Catheterisation,” *IEEE ISCAS*, 2025. **Best Demo Award.**

Additional Publications: 5 additional conference papers and live demonstrations at *IEEE ISCAS* (2025–2026), covering multimodal smart insoles, biomechanical sensing, human–machine interaction, and capsule localisation.

PATENTS

- ‘Image processing method and apparatus, electronic device and computer-readable storage medium’, CN115393922A, 2022
- ‘Data processing methods, apparatus, storage media and electronic devices’, CN114969790A, 2022
- ‘Attitude recognition method, apparatus, storage medium and electronic device’, CN114782994A, 2022
- ‘Sitting posture recognition method, device and storage medium’, CN114821652A, 2022
- ‘Face recognition method, apparatus, storage medium and electronic device’, CN114998955A, 2022
- ‘Image processing method, apparatus, storage medium and electronic device’, CN114913181A, 2022

AWARDS

- Best live demo, ISCAS, 2025
- Outstanding Employee of the Quarter in 2021, top 1.5% of 200 colleagues in the AI research institute of XDF
- Outstanding graduate from HUST in BEng and BA college in 2020
- Academic Progress Scholarship in HUST in 2019
- Outstanding Student Cadres Scholarship in HUST in 2018 and 2019
- Scientific and Technological Innovation Scholarship in HUST in 2018
- Third prize in the Mathematical Modelling Competition in central China in 2018, the top 5% of 2000 groups
- Second prize at HUST Entrepreneur Camp in 2018, second out of 26 teams

RESEARCH & PROJECTS

Research assistant: Developing the smart insole related hardware and software system	Mar 2025 – Sep 2025
Research assistant: Developing the ultra-low power embedded system for in vivo stimulation	Oct 2024 – Apr 2025
PhD research: Bioimpedance needle trajectory system	May 2024 - Oct 2024
• Conducted structural analysis of the bioimpedance needle using COMSOL Multiphysics. • Developed a complete experimental system integrating an embedded Bluetooth module, a haptic robot, and a user interface. • Key result: In 24 user trials on a venous phantom, the system achieved a 100% successful entry rate, a 0.66 mm RMS error, and an 87% probability of positioning within ± 1 mm of the vein center. The system was evaluated by NHS collaborators for potential medical needle navigation applications.	
MSc project: design and modelling of power conditioning circuits for triboelectric nanogenerator	May 2023 - Sep 2023
• Performed finite element analysis in COMSOL to model and analyze a triboelectric nanogenerator (TENG). • Designed an analytical TENG model in Verilog-AMS using Cadence. • Developed a reliable and efficient power management circuit for a bladder actuator based on the TENG models. • Key result: The system generates 3.25 mAh of energy per day with a 23% conversion efficiency, outperforming most designs.	
BEng project: design and optimisation of geomagnetic algorithm based on DSP	Feb 2020 - Jun 2020
• Optimized the World Magnetic Model 2020 (WMM2020) algorithm. • Implemented the algorithm on a DSP platform with a focus on hardware performance enhancement. • Key result: Achieved a relative error below 0.003% and a computation time of 0.57 ms, delivering twice the performance of existing systems at the time.	
Research Assistant at China National Photovoltaic Laboratory	Oct 2019 - Jan 2020
• The project, funded by Huawei, aims to measure blood flow rate and oxygen level through laser scattering imaging • Set up the camera–laser experimental platform and optimized the laser scattering imaging algorithms. • Key results: Generated thermal maps indicating blood flow rate and time-varying blood oxygen curves. Compared with existing methods, these results enable both measurements using standard mobile phones.	